MA PH 251

DIFFERENTIAL EQUATIONS FOR PHYSICS

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First-Order Ordinary Differential Equations

Chapter 1

Definition 1.1 (Order of a Differential Equation). *The order of a differential is the highest order derivative that occurs in the equation.*

- $\frac{\mathrm{d}y}{\mathrm{d}x} = -2xy$ is a first order differential equation (because the highest occurring derivative is first-order)
- $m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -kx - c\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2$ is a second order order differential equation (because it includes the second derivative, not the first derivative squared)

Definition 1.2 (Linear vs Non-linear Differential Equations). *A differential equation is "linear" if the* dependent *variable (and its derivatives) occur linearly. Otherwise it is said to be "non-linear"*

- $\frac{\mathrm{d}y}{\mathrm{d}x} = \sin(y)$ *is a non-linear differential equation because* $\sin(y)$ *is non-linear*
- $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{\mathrm{d}y}{\mathrm{d}x} + y$ *is a linear differential equation*

The most general n^{th} order linear differential equation is of the form:

$$a_{n(x)}\frac{\mathrm{d}^n y}{\mathrm{d}x^n} + a_{n-1}(x)\frac{\mathrm{d}^n y}{\mathrm{d}x^n} + ... + a_1(x)\frac{\mathrm{d}y}{\mathrm{d}x} + a_0(x)y(x) = F(x)$$

n Verifying the solution to a differential equation is as simple as taking the derivative and substituting where necessary to show that the original differential equation is satisfied. Finding the solutions is non-trivial for most types of differential equation.

Definition 1.3 (Initial Value Problem). *An initial value problem is a differential equation problem with specific values of the function (and any derivatives) at an initial point, called the "initial condition"*

- $\frac{\mathrm{d}y}{\mathrm{d}x} = y(x), y(0) = 1$ is an example of an initial value problem, outside the context of a physical application.
- Most physical applications must specify an initial condition otherwise you are said to find the "general solution"

Definition 1.4 (Boundary Value Problem). *A boundary value problem is a differential equation with a constraint on multiple values of the function (and any derivatives) for many points, called the "boundary conditions"*

- For example, a partial differential equation could specify the constraint $f(x, 0) = 2$ i.e. fixing a specific value for all values of x

Definition 1.5 (Quasi-linear First Order Differential Equation). *A differential equation of the form* $\frac{\mathrm{d}y}{\mathrm{d}x} = f(x, y)$*, i.e where the highest-order derivative is linear*

Note: A higher order differential equation can be written as a system of lower order differential equations.

$$p(x)\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+q(x)\frac{\mathrm{d}y}{\mathrm{d}x}+r(x)y=f(x)\Leftrightarrow\begin{cases}\frac{\mathrm{d}y}{\mathrm{d}x}=w\\ p(x)\frac{\mathrm{d}w}{\mathrm{d}x}+q(x)w+r(x)y=f(x)\end{cases}$$

Notice that $y(x)$ cannot be replaced in terms of w

1.1. Existence and Uniqueness of Solutions

In general and especially for physical applications, the study of differential equations is concerned about the "existence" and "uniqueness" of solutions for an equation

Equations in general can have no solutions or trivial solutions (such as when $y(x)=0$).

DEFINITION 1.6 (Unique Solution). *A solution to a differential equation is "unique" if it is the only solution to said equation.*

- $\frac{\mathrm{d}y}{\mathrm{d}x}=3y^{\frac{2}{3}}, y(0)=0$ does not have a unique solution because $y(x)=x^3$ and $y(x)=0$ are both solutions to the equation

There are a number of theorems we can use to guarantee the existence of solutions (and in some cases, at most **one** solution)

THEOREM 1.7 (Picard-Lindelof). *If $f(x,y)$ is a continuous function of x and y on a **closed** rectangle R with sides parallel to the axes and if $f(x,y)$ obeys the "Lipschitz condition" and (x_0,y_0) is an interior point in R then there is a $h>0$ such that*

$$\frac{\mathrm{d}y}{\mathrm{d}x}=f(x,y),\quad y(x_0)=y_0$$

*has exactly **one** solution.*

DEFINITION 1.8 (Lipschitz Condition). *The Lipschitz condition is defined by the following:*

$$|f(x,y_2)-(f_x,y_1)|\leq L|y_2-y_1| \text{ for some } L, 0\leq L<\infty$$

DEFINITION 1.9 (Picard). *If $f(x,y)$ is a continuous function of x and y on a **closed** rectangle R with sides parallel to the axes and if $\frac{\delta f}{\delta y}$ is continuous and (x_0,y_0) is an interior point in R then there is a $h>0$ such that*

$$\frac{\mathrm{d}y}{\mathrm{d}x}=f(x,y),\quad y(x_0)=y_0$$

*has exactly **one** solution.*

A proof of the Theorem 1.7 can be achieved using the Banach Contraction Mapping Principal, for worked-through version of the proof, see the written notes.

However, the solution for *some* differential equations can be approximated using the iterative technique of "Picard iterations", For a differential equation $\frac{\mathrm{d}y}{\mathrm{d}x}=f(y,x)$ begin with some initial function y_0 typically $y_0(x)=0$ and define the next iteration to be,

$$y_{n+1}=y_0+\int_{x_0}^{x}f(s,y_n(s))\,\mathrm{d}s$$

1.2. Separable Differential Equations

Definition 1.10 (Separable Differential Equation). *A differential equation is called separable if it can be written in the form* $\frac{\mathrm{d}y}{\mathrm{d}x} = g(x)h(y)$.

A separable differential equation can be solved trivially by isolating the dependent and independent variables are separate sides of the equation and performing two integrals.

Proof. Suppose $\frac{\mathrm{d}y}{\mathrm{d}x} = g(x)h(y)$ where $h(y_0) \neq 0$ and $H(y) \equiv \int_{y_0}^{y} \frac{\mathrm{d}y}{h(y)}$

Using a substitution of variables where $u = y(x)$ $\frac{\mathrm{d}u}{\mathrm{d}x} = y'(x)$ we can rewrite the integral

$$\begin{aligned}\int_{y_0}^{y} \frac{\mathrm{d}y}{h(y)} &= \int_{x_0}^{x} \frac{1}{h(y(x))} y'(x)\,\mathrm{d}x \\ &= \int_{x_0}^{x} \frac{1}{h(y(x))} g(x)h(y)\,\mathrm{d}x \\ &= \int_{x_0}^{x} g(x)\,\mathrm{d}x\end{aligned}$$

Note, solving separable differential equations often requires skills of integration including partial fraction decomposition, trigonometric substitution, substitution of variables, integration by parts, etc.

Here is a simplified overview of all of these techniques

1.2.1. Review of Integration Techniques

- Integration by Parts: $\int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}v$
  - For most cases, identify a term (in a product) that may simplify when differentiated as u
- Trigonometric Substitution: Set $x = \sin(\theta)$, $x = \tan(\theta)$ or $x = \sec(\theta)$
  - For when the integrand includes a radical containing the sum or difference of two squares
- Trigonometric Integrals: A special case of a substitution of variables
  - For when the integrand contains powers of trigonometric functions
  - $$\int \sin^m(x)\cos^n(x)\,\mathrm{d}x \Rightarrow \begin{cases} \text{n is odd} \to u = \sin(x) \\ \text{m is even} \to u = \cos(x) \\ \text{n and m are even} \to \sin(2x) = 2\sin(x)\cos(x) \end{cases}$$
  - $$\int \tan^m(x)\sec^{n(x)}\,\mathrm{d}x \Rightarrow \begin{cases} \text{n is even} \to u = \tan(x) \\ \text{m is odd and n} \geq 1 \to u = \sec(x) \\ \text{otherwise, use other techniques} \end{cases}$$
- A fraction of the form $\frac{N(x)}{D(x)}$ can be simplified using partial fraction decomposition.
  - If $\deg(N(x)) > \deg(D(x))$ divide and then simplify
  - Otherwise, split D into different factors and divide the numerator into some known form
    - $D(x) = ax + b \to \frac{A}{ax+b}$
    - $D(x) = ax^2 + bx + c \to \frac{Ax+B}{ax^2+bx+c}$
    - $D(x) = (ax+b)^n \to \frac{A_1}{(ax+b)^n} + ... + \frac{A_n}{ax+b}$
- Knowing a number of common integrals is also important, this includes trigonometric, hyperbolic, inverse trigonometric, logarithmic, etc.

1.3. Differential Equations of Homogeneous Functions

Definition 1.11 (Differential Form). *The differential form for a quasilinear first order differential equation:*

$$M(x,y)\,\mathrm{d}x + N(x,y)\,\mathrm{d}y = 0$$

Definition 1.12 (Homogeneous Function). *A function f is homogeneous of degree n if and only if the following holds*

$$f(tx, ty, ...) = t^n f(x, y, ...)$$

A differential equation with M and N both homogeneous with degree n can be written as

$$\frac{\mathrm{d}y}{\mathrm{d}x} = f(x,y) = -\frac{M(x,y)}{N(x,y)}$$

where f is now homogeneous of degree 0. This being the case, we can make the substitution $z \equiv \frac{y}{x}$, turning the bivariate function f into a single variable function $f(1,z)$

It also follows from differentiating z that $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}z}{\mathrm{d}x}x + z$, ultimately transforming the differential equation into the separable form

$$\frac{\mathrm{d}z}{f(1,z) - z} = \frac{\mathrm{d}x}{x}$$

1.4. Exact Differential Equations

Consider a smooth relation of the form $f(x,y) = c$, we can implicitly differentiate this relation to obtain the equation

$$\frac{\mathrm{d}}{\mathrm{d}x}(f(x,y)) = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}x} = 0$$

which takes a similar form to the differential form. This begs the question whether we are able to find a function $f(x,y)$ such that $\frac{\partial f}{\partial x} = M(x,y)$ and $\frac{\partial f}{\partial y} = N(x,y)$

$$M(x,y)\,\mathrm{d}x + N(x,y)\,\mathrm{d}y = 0 \text{ is exact} \rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \text{ on an open region, } D$$

The converse requires a slightly stricter restriction on the domain. The converse requires $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$ are continuous everywhere on a simply connected domain.

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \rightarrow M(x,y)\,\mathrm{d}x + N(x,y)\,\mathrm{d}y = 0 \text{ is exact}$$

A simply connected domain is one in which any curve can be contracted continuously into a single point without leaving the domain. (No holes in the domain)

To solve an exact differential equation we can consume one of our functions M or N to find an answer through partial integration

$$f(x,y) = \int_{x_0}^{x} M\,\mathrm{d}x + \varphi(y)$$

$$\frac{\partial f}{\partial y} = N(x,y) \Rightarrow \int_{y_0}^{y}\int_{x_0}^{x} M\,\mathrm{d}x + \varphi'(y)\,\mathrm{d}y = N(x,y)$$

We hold on to the belief that $\varphi'(y)$ will be a function of only y after algebraic elimination.

1.5. Integrating Factors

Exact equations are difficult to come by, luckily we can find attempt to find an integrating factor, $\mu(x, y)$ such that

$$\frac{\partial(\mu M)}{\partial y} = \frac{\partial(\mu N)}{\partial x}$$

Using implicit differentiation, we can attempt to isolate for μ

$$\frac{\partial(\mu M)}{\partial y} = \frac{\partial(\mu N)}{\partial x} \to \frac{\partial \mu}{\partial y}M + \mu\frac{\partial M}{\partial y} = \frac{\partial \mu}{\partial x}N + \mu\frac{\partial N}{\partial x} \to \mu\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) = N\frac{\partial \mu}{\partial x} - M\frac{\partial \mu}{\partial y}$$

At this point, solving this linear first-order partial differential equation is just as difficult as the quasi-linear first order differential equation. However, we can attempt to guess at solutions by taking μ as a function of x or y alone.

Once we have our integrating factor, we can solve for the solution using the same strategy as exact expression.

1.6. Linear Differential Equations

First order linear differential equations are of the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$$

These equations are a special case of the integration factor, where $\mu = e^{\int P(x)\,\mathrm{d}x}$.

Notice that after applying the factor, the term including the first derivative and linear term take the form of the result of applying the product rule. This allows us to solve for y explicitly as a function of x